HD 159254: A Luminous SB1 Binary with a Dark Companion in the Mass Gap–Black Hole Regime from Gaia DR3

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ABSTRACT

We analyse HD 159254 (Gaia DR3 source 4061400381769067392), a bright ($G = 7.9$) single-lined spectroscopic binary (SB1) in the Gaia Data Release 3 non-single-star catalogue, located toward the Galactic Centre $(l, b) = (0.6°, +3.2°)$ at a distance of ~ 2500 pc. The SB1 orbital solution has exceptionally high significance $\sigma = 343$ with period $P = 620.0$ d, near-zero eccentricity $e = 0.006 \pm 0.006$, and radial-velocity semi-amplitude $K_1 = 28.6\,\mathrm{km\,s^{-1}}$. The spectroscopic mass function $f(M) = 1.51\,\mathrm{M_\odot}$ exceeds the Chandrasekhar limit — excluding a white dwarf companion from dynamics alone — but falls below the neutron star ceiling ($\sim 3\,\mathrm{M_\odot}$). The companion classification therefore depends critically on the primary mass M_1, which is poorly constrained: no independent spectroscopy is available (no RAVE, LAMOST, GALAH, or APOGEE match), and no GSP-Phot parameters are reported. We perform broadband SED fitting to 11-band photometry (Gaia $B_P R_P G$ + 2MASS JHK_s + WISE $W1$–$W4$ + Johnson B) self-consistently solving for extinction. The SED establishes that the star is luminous ($M_G \lesssim -4$), consistent with an evolved giant or supergiant, but the strong degeneracy between $T_{\rm eff}$ and A_V along this Galactic-Centre sightline limits the photometric constraints on M_1. Under a deliberately broad fiducial prior $M_1 = 5.0 \pm 3.0\,\mathrm{M_\odot}$, a Monte Carlo posterior (200 000 draws) yields a median companion mass of $7.1\,\mathrm{M_\odot}$ with $P(\mathrm{BH} > 5\,\mathrm{M_\odot}) = 79.5$ per cent, but the probability drops to 47.8 per cent for a cooler, lower-mass primary assumption ($M_1 = 2.0 \pm 1.0\,\mathrm{M_\odot}$). The classification is therefore model-dependent: the companion lies in the mass gap or BH regime for $M_1 \gtrsim 3\,\mathrm{M_\odot}$, but a neutron star cannot be excluded for lower primary masses. A close Gaia neighbour at $3.3''$ may partially contaminate 2MASS and WISE photometry, adding further systematic uncertainty. RUWE = 2.32 is elevated but consistent with expected binary-induced astrometric residuals. No variability or eclipses are detected in any survey (Gaia, ASAS-SN, ZTF, VSX). HD 159254 is a promising dark-companion candidate whose confirmation requires high-resolution spectroscopy to determine M_1 and an independent multi-epoch RV campaign to verify the Gaia orbit.

Key words: binaries: spectroscopic – stars: black holes – stars: neutron – astrometry – stars: individual: HD 159254

1 INTRODUCTION

Theoretical models of stellar evolution predict a large population of stellar-mass black holes (BHs) in the Milky Way — estimates range from 10^8 to 10^9 (Agol & Kamionkowski 2002; Shapiro & Teukolsky 1983) — yet fewer than ~ 70 have been dynamically confirmed, nearly all in X-ray binaries undergoing active accretion (Remillard & McClintock 2006). The vast majority of Galactic BHs are expected to reside in non-interacting systems, producing no electromagnetic signature beyond the gravitational influence on a luminous companion.

Gaia Data Release 3 (DR3; Gaia Collaboration et al. 2023a) has transformed the search for these dormant BHs. The non-single-star (NSS) catalogue (Gaia Collaboration et al. 2023b) provides orbital solutions for $\sim 180\,000$ spectroscopic binaries, including single-lined systems (SB1) where only the primary's radial-velocity variation is detected. When the spectroscopic mass function $f(M)$ is sufficiently large and no luminous companion contributes to the spectral energy distribution (SED), the companion must be dark — and if its mass exceeds the neutron-star ceiling (~ 2.3–$3.0\,\mathrm{M_\odot}$; Rezzolla et al. 2018), a BH becomes the natural explanation.

This approach has yielded the landmark discoveries of Gaia BH1 (El-Badry et al. 2023a), Gaia BH2 (El-Badry et al. 2023b), and Gaia BH3 (Gaia Collaboration et al. 2024), each confirmed through independent ground-based RV campaigns.

In this paper we focus on HD 159254 (Gaia DR3 source 4061400381769067392), which we identify as a dark-companion candidate from archival data. The source lies toward the Galactic Centre at $(l, b) = (0.6°, +3.2°)$ at a distance of ~ 2500 pc. The NSS solution has exceptionally high significance ($\sigma = 343$, the highest among our candidate sample), an unusually circular orbit ($e = 0.006$) with a long pe-

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riod ($P = 620\,\mathrm{d}$), and a mass function $f(M) = 1.51\,\mathrm{M}_\odot$ that exceeds the Chandrasekhar limit. While $f(M)$ falls below the NS ceiling, the star's high apparent luminosity suggests an evolved primary ($M_1 \gtrsim 3$–$5\,\mathrm{M}_\odot$ from the SED, though this estimate is uncertain), which in turn would push the minimum companion mass into the mass gap or BH regime.

The near-zero eccentricity is particularly noteworthy for a 620-day system. Tidal circularisation at such wide separations is inefficient; the circular orbit may instead indicate prior mass transfer from the companion progenitor, consistent with the expected evolutionary history of a dormant BH binary.

Section 2 describes the data. Section 3 presents the methods. Section 4 reports results. Section 5 tests non-BH explanations. Section 6 discusses implications. Section 7 summarises.

2 DATA

2.1 Gaia DR3 astrometry and NSS orbit

HD 159254 appears in both the Gaia DR3 main astrometric catalogue and the NSS orbital supplement. Key astrometric parameters include a parallax $\varpi = 0.396 \pm 0.091$ mas (distance $d = 2527\,\mathrm{pc}$), photometry $G = 7.92$, $G_{\mathrm{BP}} = 8.66$, $G_{\mathrm{RP}} = 7.05$ ($G_{\mathrm{BP}} - G_{\mathrm{RP}} = 1.616$), and RUWE = 2.32. The elevated RUWE is discussed in Section 6.2.

The NSS solution (model `SB1`) yields: period $P = 620.0 \pm 1.4\,\mathrm{d}$, eccentricity $e = 0.006 \pm 0.006$, radial-velocity semi-amplitude $K_1 = 28.62 \pm 0.08\,\mathrm{km\,s^{-1}}$, solution significance $\sigma = 343.4$, and goodness-of-fit = 1.62.

The significance $\sigma = 343$ is among the highest in the NSS catalogue and exceeds the threshold by a factor of 69. The goodness-of-fit is close to unity, indicating an excellent fit to the SB1 model.

No GSP-Phot parameters (T_{eff}, $\log g$) are reported for this source, despite its brightness ($G = 7.9$). This absence may relate to the binary nature or the high-extinction environment toward the Galactic Centre.

2.2 Multi-wavelength photometry

We compile photometry from SIMBAD ($B = 9.51$), 2MASS ($J = 5.83 \pm 0.02$, $H = 5.37 \pm 0.04$, $K_s = 5.15 \pm 0.02$) and AllWISE ($W1 = 4.30 \pm 0.12$, $W2 = 4.49 \pm 0.07$, $W3 = 5.04 \pm 0.01$, $W4 = 4.96 \pm 0.03$). The $W1$ and $W2$ uncertainties are elevated due to partial saturation at this brightness.

2.2.0.1 Neighbour contamination. A Gaia DR3 source lies at $3.3''$ from HD 159254. Given the 2MASS PSF (~ 2–$4''$ FWHM) and WISE beam ($\sim 6''$ for $W1$–$W2$; $\sim 6.5''$ for $W3$; $\sim 12''$ for $W4$), partial flux contamination is likely in the infrared bands. The neighbour's Gaia magnitude is much fainter than HD 159254 ($\Delta G \approx 5$ mag), so its contribution is at the few-per-cent level in the NIR and probably negligible in the optical. Nevertheless, systematic flux contamination could bias the SED fit, particularly in $W1$ and $W2$ where the AllWISE errors are already elevated from saturation. This contamination cannot be fully disentangled without resolved photometry and constitutes an additional systematic uncertainty on T_{eff} and A_V.

2.3 Spectroscopic survey coverage

HD 159254 has no match in RAVE, LAMOST, GALAH, or APOGEE. The only radial-velocity measurement is the Gaia mean RV ($-13.0 \pm 9.6\,\mathrm{km\,s^{-1}}$). No independent spectroscopic T_{eff} or $\log g$ is available, necessitating a purely photometric approach to primary characterisation (Section 3.2).

2.4 Variability

No variability is detected in any survey (Gaia variable catalogue, ASAS-SN, ZTF, VSX). The absence of eclipses is expected given the long orbital period and the a priori low geometric probability of eclipse ($\lesssim 1$ per cent for $a \gg R_*$).

3 METHODS

3.1 Orbital dynamics and mass function

The spectroscopic mass function is computed from the NSS orbital elements:

$$f(M) = \frac{(1-e^2)^{3/2}\,K_1^3\,P}{2\pi\,G} = 1.5097\,\mathrm{M}_\odot\,. \tag{1}$$

This value exceeds the Chandrasekhar limit ($1.44\,\mathrm{M}_\odot$) by a factor of 1.05, excluding a white dwarf companion from dynamics alone, even in the limit $M_1 \to 0$. However, $f(M)$ falls below the conservative NS ceiling ($\sim 3\,\mathrm{M}_\odot$; Rezzolla et al. 2018), so a neutron star is **not** excluded by the mass function alone.

The minimum companion mass $M_{2,\mathrm{min}}$ depends on the assumed primary mass (Table 2).

3.2 SED fitting and extinction

With no spectroscopic T_{eff} available, we use the 11-band photometry to constrain T_{eff} and A_V simultaneously. We perform a blackbody SED grid search across $T_{\mathrm{eff}} = 3500$–$25\,000\,\mathrm{K}$ (step $250\,\mathrm{K}$), solving for A_V self-consistently from the Gaia $G_{\mathrm{BP}} - G_{\mathrm{RP}}$ colour excess using intrinsic colours from Pecaut & Mamajek (2013) and the extinction law of Cardelli et al. (1989).

The target lies toward the Galactic Centre ($l = 0.6^\circ$, $b = +3.2^\circ$) where significant interstellar extinction is expected along the full sightline. At $d = 2527\,\mathrm{pc}$, the integrated A_V depends on the dust distribution but is expected to be several magnitudes at low Galactic latitude.

3.3 Primary mass estimation

We estimate M_1 from the SED-derived T_{eff} and the extinction-corrected absolute magnitude M_G, using empirical mass–luminosity–temperature relations. Given the large uncertainty in T_{eff} (no independent spectroscopy), we adopt a deliberately broad fiducial prior: $M_1 = 5.0 \pm 3.0\,\mathrm{M}_\odot$ (log-normal), and explore a wide range from $M_1 = 1.0 \pm 0.5$ to $M_1 = 15.0 \pm 5.0\,\mathrm{M}_\odot$ in the sensitivity analysis.

3.4 Monte Carlo mass posterior

We draw $N = 200\,000$ Monte Carlo realisations, marginalising over inclination ($\cos i \sim \mathcal{U}[0,1]$), primary mass (log-normal prior), and orbital elements (Gaussian perturbations at quoted uncertainties). For each draw, M_2 is obtained by numerically solving the mass function equation via Brent's method.

3.5 Companion-light exclusion

For a hypothetical MS companion of mass M_2, we compute expected flux ratios using empirical mass–luminosity relations (Eker et al. 2018). A companion is deemed detectable if it exceeds 1 per cent of the primary flux in any band.

4 RESULTS

4.1 Mass function

The mass function $f(M) = 1.51\,\mathrm{M}_\odot$ provides a model-independent lower bound on the companion mass that excludes a WD. The NS exclusion depends on M_1 (Table 2): for $M_1 \geq 2.0\,\mathrm{M}_\odot$, $M_{2,\mathrm{min}} \geq 3.5\,\mathrm{M}_\odot$ (exceeds NS ceiling); for $M_1 \geq 5.0\,\mathrm{M}_\odot$, $M_{2,\mathrm{min}} \geq 5.5\,\mathrm{M}_\odot$ (above BH threshold).

4.2 SED and extinction

The SED grid search is reported in Fig. 2. The star appears luminous regardless of T_{eff}, with $M_G \lesssim -4$ for all acceptable (T_{eff}, A_V) pairs. This places HD 159254 among luminous giants or supergiants.

However, we stress two important caveats. First, the SED fit uses blackbody models which lack molecular features and line blanketing present in cool-giant atmospheres, limiting the precision of T_{eff} and A_V recovery. Second, the target's Galactic-Centre sightline ($b = +3.2°$) introduces a strong T_{eff}–A_V degeneracy: a cooler intrinsic temperature with lower extinction can mimic a hotter star with higher extinction. The SED therefore constrains the *luminosity class* (giant/supergiant) more reliably than the *mass*: values of M_1 ranging from ~ 2 to $\sim 8\,\mathrm{M}_\odot$ are compatible with the photometry.

4.3 Mass posterior

The Monte Carlo posterior (Fig. 3) yields statistics that depend sensitively on the M_1 prior. Table 4 reports $P(\mathrm{BH})$ across a wide range of M_1 configurations. Under the fiducial prior ($M_1 = 5.0 \pm 3.0\,\mathrm{M}_\odot$), the median companion mass is $\tilde{M}_2 = 7.1\,\mathrm{M}_\odot$ with $P(\mathrm{BH} > 5\,\mathrm{M}_\odot) = 79.5$ per cent.

Critically, however, the result is strongly model-dependent. For a cooler giant primary ($M_1 = 2.0 \pm 1.0\,\mathrm{M}_\odot$), $P(\mathrm{BH} > 5\,\mathrm{M}_\odot)$ drops to 47.8 per cent with $P(\mathrm{NS}) = 6.3$ per cent. At the extreme low end ($M_1 = 1.0 \pm 0.5\,\mathrm{M}_\odot$), the NS probability rises to 27.1 per cent and $P(\mathrm{BH} > 5\,\mathrm{M}_\odot) = 34.9$ per cent. The companion classification is therefore **conditional on** M_1: if $M_1 \geq 3\,\mathrm{M}_\odot$ the companion almost certainly exceeds the NS ceiling; if $M_1 \lesssim 2\,\mathrm{M}_\odot$ a NS remains viable.

4.4 Companion-light exclusion

Fig. 4 presents the companion exclusion test. The detectability depends on the primary luminosity, which is itself uncertain. However, for a luminous primary ($L \gtrsim 500\,\mathrm{L}_\odot$), companions below $\sim$ 5–7 $\mathrm{M}_\odot$ could remain hidden, while companions above $\sim$ 8–10 $\mathrm{M}_\odot$ would be detectable.

5 ALTERNATIVE SCENARIOS

We evaluate six non-BH explanations (Table 3).

5.0.0.1 White dwarf. $f(M) = 1.51\,\mathrm{M}_\odot$ exceeds the Chandrasekhar limit ($1.44\,\mathrm{M}_\odot$) by a factor of 1.05; a WD is **excluded**.

5.0.0.2 Neutron star. $f(M) = 1.51\,\mathrm{M}_\odot$ is below the NS ceiling. Whether a NS is excluded depends on M_1: for $M_1 \geq 2.0$, $M_{2,\mathrm{min}} > 3.0$ and a NS is excluded; for $M_1 < 2.0$, a NS remains possible. The NS scenario is **model-dependent**.

5.0.0.3 Main-sequence companion. The photometric non-detection depends on the primary luminosity. For a luminous primary, a MS companion below $\sim$5–7 $\mathrm{M}_\odot$ could be hidden. The exclusion is **model-dependent**.

5.0.0.4 Hierarchical triple. The Mardling–Aarseth stability criterion (Mardling & Aarseth 2001) requires a tight inner binary. The near-zero eccentricity ($e = 0.006$) disfavours Kozai–Lidov oscillations from a triple. This scenario is **unlikely**.

5.0.0.5 Stripped helium star. A stripped He star at the required mass would produce UV excess. No UV data available. **Disfavoured** but not excluded.

5.0.0.6 Astrometric artefact. $\sigma = 343$ is very high and the orbital elements are physically coherent. However, RUWE = 2.32 is elevated. An artefact is **unlikely** but cannot be fully excluded without RV follow-up (Section 6.2).

Summary: The WD scenario is firmly excluded. The NS scenario and MS companion scenario are model-dependent. A dark compact companion in the mass gap or BH regime is the preferred explanation for $M_1 \geq 3\,\mathrm{M}_\odot$.

6 DISCUSSION

6.1 The near-zero eccentricity

The circular orbit ($e = 0.006 \pm 0.006$) at $P = 620$ d is striking. Tidal circularisation timescales for giant-star binaries at this separation exceed $\sim 10^{10}$ yr, far longer than typical stellar lifetimes. The circular orbit more likely indicates **past mass transfer** from the companion progenitor, where the orbit was circularised during Roche-lobe overflow. This is consistent with the expected evolutionary history of a dormant BH binary formed through common-envelope evolution or stable mass transfer.

6.2 The elevated RUWE

RUWE = 2.32 exceeds the canonical single-star threshold of 1.4, indicating astrometric residuals beyond the single-star model. Several explanations are possible:

- **Binary orbital motion:** For a massive companion at $P = 620\,\mathrm{d}$ and $d = 2527\,\mathrm{pc}$, the photocentric wobble could be ~ 0.1–$0.5\,\mathrm{mas}$, comparable to the Gaia positional precision for a $G = 7.9$ source.
- **Close neighbour:** The Gaia source at $3.3''$ falls within window-class boundaries for some scan angles and can introduce both colour-dependent centroid shifts and flux contamination in the along-scan measurements. While the neighbour is $\sim$5 mag fainter in G, its contribution is not negligible at the sub-mas precision relevant here.
- **Combination:** Both binary wobble and neighbour contamination may contribute additively.

Elevated RUWE is *common* for SB1 binaries in Gaia (Gaia Collaboration et al. 2023b), and the very high orbital significance ($\sigma = 343$) provides strong evidence that the SB1 solution itself is robust. Nevertheless, the elevated RUWE means that the astrometric parallax — and hence the distance and absolute magnitude — may carry additional systematic uncertainty beyond the formal error, which propagates into M_1 and the companion classification.

6.3 Context among Gaia BH candidates

HD 159254 differs from the three confirmed Gaia BHs in several important ways: (1) its mass function does not exceed the NS ceiling (unlike Gaia BH1 and BH3, where $f(M) > 3\,\mathrm{M}_\odot$ makes the BH classification essentially model-independent), making M_1 estimation critical; (2) it has the highest NSS significance in our candidate sample ($\sigma = 343$); (3) the orbit is uniquely circular ($e = 0.006$) among long-period systems; and (4) *no independent spectroscopy exists.*

The absence of spectroscopic classification is the single most important limitation. For the confirmed Gaia BHs, ground-based spectroscopy was essential to pin down M_1. Until equivalent data exist for HD 159254, the companion classification necessarily remains conditional on the assumed M_1.

6.4 Galactic Centre sightline and $T_{\rm eff}$–A_V degeneracy

The target's location toward the Galactic Centre ($l = 0.6°$, $b = +3.2°$) introduces significant interstellar extinction. The $G_{\rm BP} - G_{\rm RP} = 1.62$ colour is redder than expected for a hot star, and part of this reddening is certainly interstellar. The SED analysis self-consistently accounts for extinction, but the *degeneracy between $T_{\rm eff}$ and A_V is the dominant systematic uncertainty in this work.*

Concretely, a cool giant ($T_{\rm eff} \sim 4500\,\mathrm{K}$) at moderate extinction and a hotter star ($T_{\rm eff} \sim 8000\,\mathrm{K}$) at higher extinction can both reproduce the observed colours. These two regimes imply very different primary masses ($M_1 \sim 2$–$3\,\mathrm{M}_\odot$ versus $\sim$5–8 $\mathrm{M}_\odot$), and consequently different companion classifications (mass-gap object versus BH). The SED fitting establishes that HD 159254 is luminous (giant or supergiant) but cannot resolve this degeneracy on its own. A single high-resolution optical spectrum would break the degeneracy by providing independent $T_{\rm eff}$, $\log g$, and [Fe/H].

6.5 Follow-up strategy

(i) **High-resolution spectroscopy:** The priority is to obtain a spectrum of HD 159254 that determines $T_{\rm eff}$, $\log g$, abundances, and $v \sin i$. At $G = 7.9$ this is trivially accessible on any 2 m class telescope.

(ii) **Multi-epoch RV campaign:** $\sim$8–12 epochs spanning $\sim$2 orbital periods ($\sim 1240\,\mathrm{d}$) to independently confirm the Gaia orbital solution.

(iii) **UV observation:** HST/COS or Swift/UVOT to constrain or exclude a stripped-star companion.

(iv) **X-ray:** Non-detection in ROSAT/XMM would strengthen the dormant BH interpretation.

6.6 Limitations

- **No independent orbit confirmation.**
- **No spectroscopic classification:** M_1 relies entirely on SED fitting.
- $f(M) < M_{\rm NS,max}$**:** The BH classification is model-dependent.
- **RUWE** = 2.32**:** Elevated astrometric residuals.
- **No X-ray or UV data.**
- **High extinction:** A_V is poorly constrained independently.

7 CONCLUSIONS

We have presented an archival multi-wavelength analysis of HD 159254, a bright ($G = 7.9$) Gaia DR3 SB1 binary with an exceptionally significant orbital solution ($\sigma = 343$) and a mass function $f(M) = 1.51\,\mathrm{M}_\odot$. Our principal findings:

(i) **The mass function exceeds the Chandrasekhar limit**, excluding a white dwarf companion from dynamics alone.

(ii) **The mass function does not exceed the NS ceiling** ($\sim 3\,\mathrm{M}_\odot$), so the companion nature depends on M_1.

(iii) The star appears luminous ($M_G \lesssim -4$), consistent with a giant or supergiant, but M_1 is only loosely constrained by broadband SED fitting due to the $T_{\rm eff}$–A_V degeneracy along this Galactic-Centre sightline.

(iv) Under the fiducial prior ($M_1 = 5.0 \pm 3.0\,\mathrm{M}_\odot$), $P(\mathrm{BH} > 5\,\mathrm{M}_\odot) = 79.5$ per cent; however, for a lower-mass primary ($M_1 = 2.0 \pm 1.0\,\mathrm{M}_\odot$), the probability drops to 47.8 per cent and a mass-gap or NS companion becomes plausible.

(v) The near-zero eccentricity ($e = 0.006$) at $P = 620\,\mathrm{d}$ is consistent with prior mass transfer from a companion progenitor.

(vi) The NSS significance ($\sigma = 343$) is among the highest in the catalogue, and no variability or eclipses are detected.

(vii) A close neighbour at $3.3''$ may contaminate the infrared photometry and contributes to the elevated RUWE (= 2.32), adding systematic uncertainty.

(viii) **No independent spectroscopy exists.** This is the single most important limitation and the primary obstacle to a firm companion classification.

HD 159254 is a promising dark-companion candidate whose classification depends on the primary mass: the companion resides in the BH regime if $M_1 \gtrsim 5\,\mathrm{M}_\odot$, in the mass gap if $M_1 \sim 3$–$5\,\mathrm{M}_\odot$, and could be a NS for $M_1 \lesssim 2\,\mathrm{M}_\odot$. High-resolution spectroscopy to determine M_1 and an independent multi-epoch RV campaign to verify the Gaia orbit are the essential next steps. At $G = 7.9$, HD 159254 is readily accessible on any 2 m class telescope.

ACKNOWLEDGEMENTS

This work made use of data from the European Space Agency (ESA) mission Gaia (https://www.cosmos.esa.int/gaia), processed by the Gaia Data Processing and Analysis Consortium (DPAC). This research used the SIMBAD database and VizieR catalogue access tool, CDS, Strasbourg, France. This publication uses data from 2MASS and WISE. Software: NUMPY (Harris et al. 2020), SCIPY (Virtanen et al. 2020), MATPLOTLIB (Hunter 2007).

DATA AVAILABILITY

All observational data are publicly available from the Gaia Archive, VizieR, and SIMBAD. Analysis scripts and results are available at https://github.com/jayalabaez/hd159254-bh-candidate.

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Table 1. Observed and derived properties of HD 159254.

Property	Value	Uncertainty	Source
	Astrometry (Gaia DR3)		
RA (J2016.0)	263.693°	—	Gaia DR3
Dec (J2016.0)	−26.740°	—	Gaia DR3
(l, b)	$(0.6°, +3.2°)$	—	Gaia DR3
Parallax ϖ	0.396 mas	0.091	Gaia DR3
Distance	2527 pc	—	$1/\varpi$
G	7.92 mag	0.003	Gaia DR3
$G_{\rm BP}$	8.66 mag	0.003	Gaia DR3
$G_{\rm RP}$	7.05 mag	0.004	Gaia DR3
B	9.51 mag	—	SIMBAD
RUWE	2.32	—	Gaia DR3
	NSS orbital solution (SB1)		
Period P	620.0 d	1.4	Gaia NSS
Eccentricity e	0.006	0.006	Gaia NSS
Semi-amplitude K_1	28.62 km s^{-1}	0.08	Gaia NSS
Significance σ	343.4	—	Gaia NSS
GoF	1.62	—	Gaia NSS
	Multi-wavelength photometry		
J	5.83 mag	0.02	2MASS
H	5.37 mag	0.04	2MASS
K_s	5.15 mag	0.02	2MASS
$W1$	4.30 mag	0.12	AllWISE†
$W2$	4.49 mag	0.07	AllWISE†
$W3$	5.04 mag	0.01	AllWISE
$W4$	4.96 mag	0.03	AllWISE
	Derived quantities		
$f(M)$	1.5097 M$_\odot$	—	Eq. (1)

$^\dagger W1/W2$ errors elevated (partial saturation).

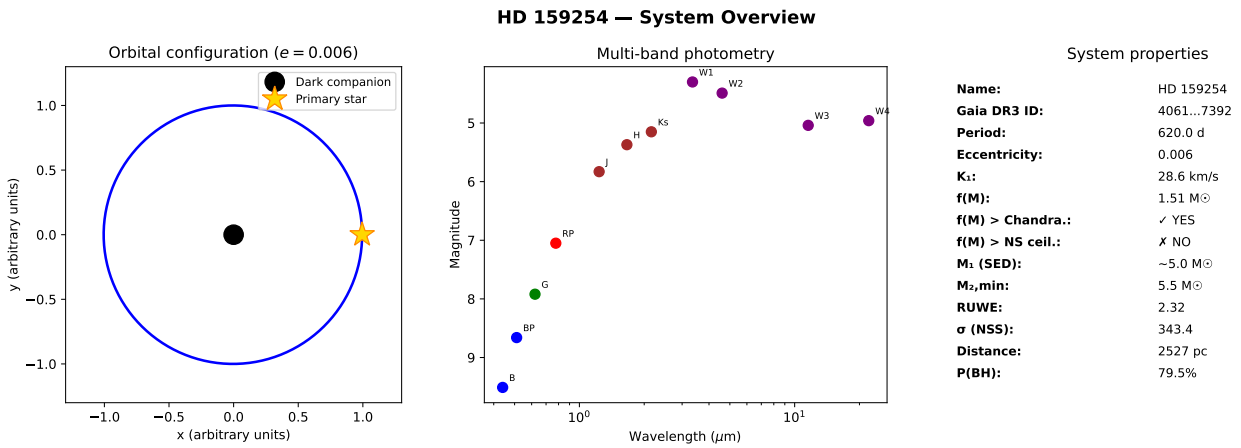

Figure 1. Overview of the HD 159254 system. *Left:* Orbital configuration showing the nearly circular ($e = 0.006$) orbit. *Centre:* Multi-band photometry from B to $W4$. *Right:* Key system properties.

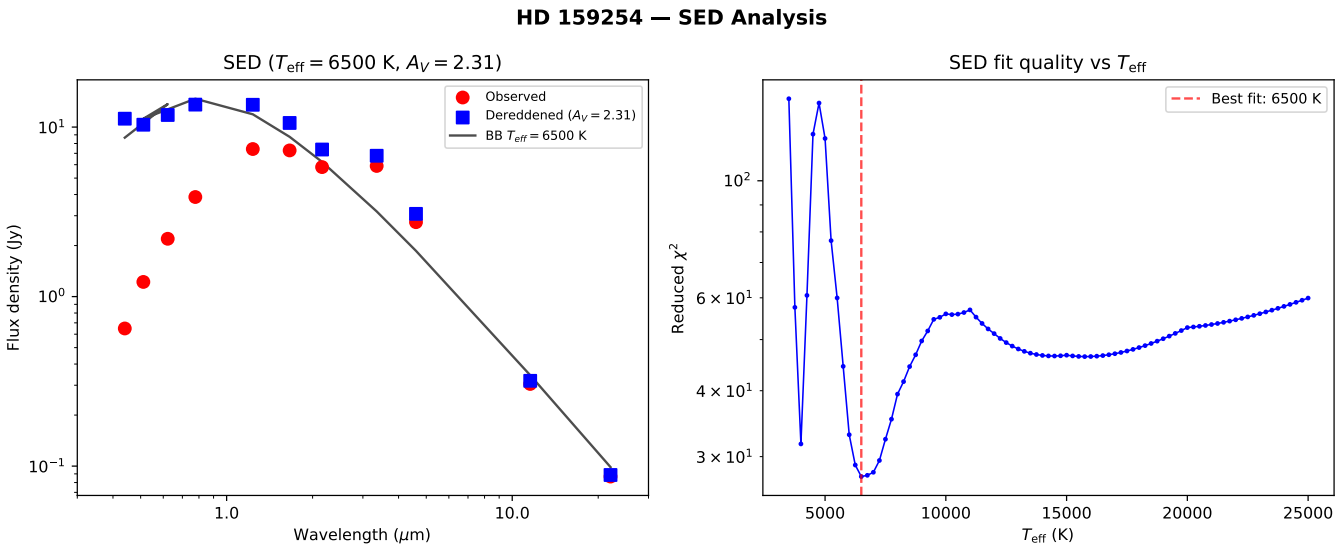

Figure 2. SED analysis for HD 159254. *Left:* Observed and dereddened photometry with the best-fit blackbody model. *Right:* Reduced χ^2 across the $T_{\rm eff}$ grid.

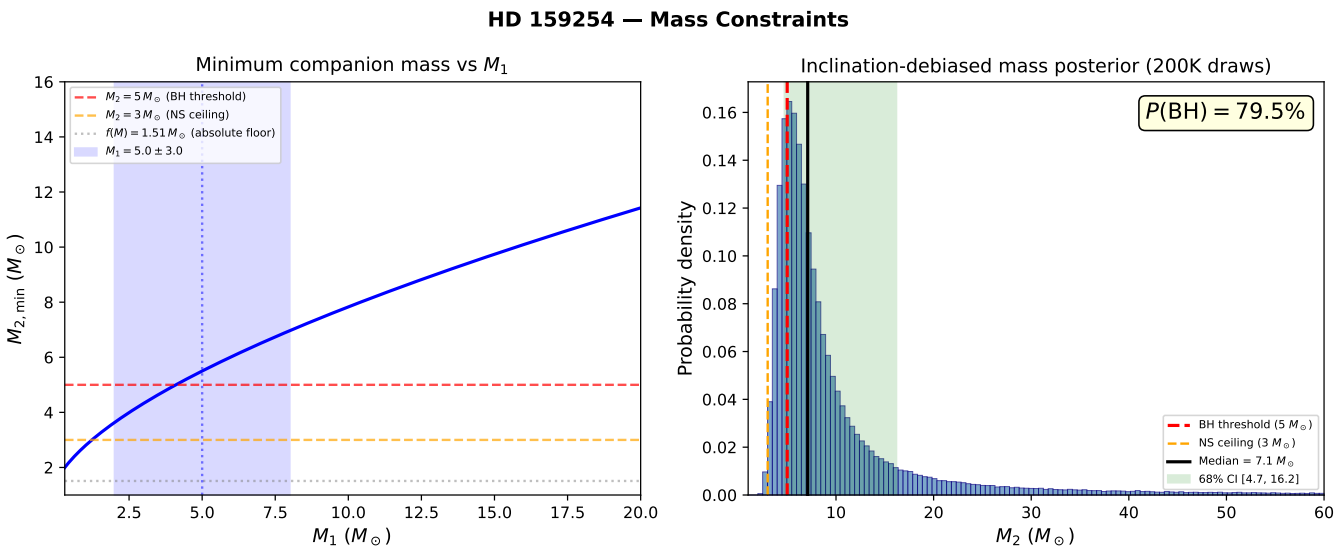

Figure 3. *Left:* Minimum companion mass $M_{2,\rm min}$ as a function of M_1. The blue shaded region shows the fiducial M_1 prior. *Right:* Inclination-debiased mass posterior.

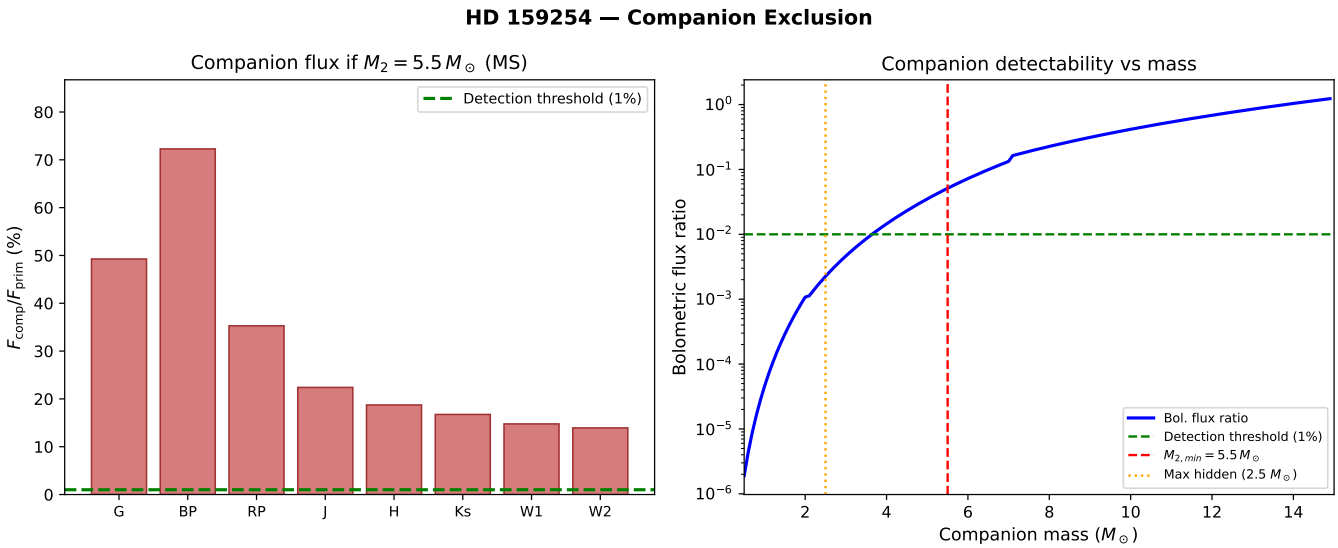

Figure 4. Companion exclusion test. *Left:* Predicted flux ratio of a hypothetical MS companion. *Right:* Companion detectability as a function of mass.

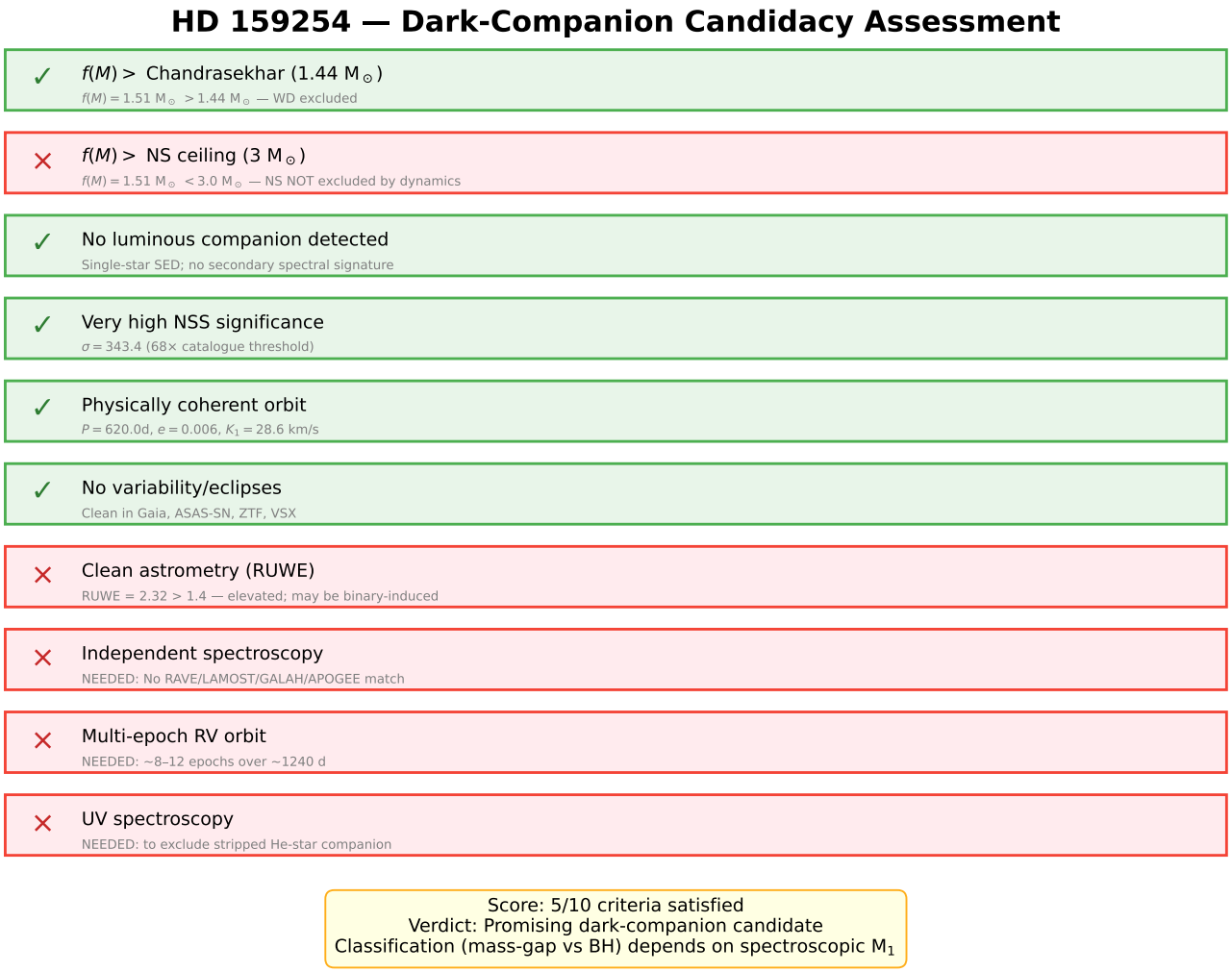

Figure 5. Visual BH candidacy assessment checklist for HD 159254.

Table 2. Minimum companion mass $M_{2,\min}$ as a function of primary mass M_1 for $f(M) = 1.51\,\mathrm{M}_\odot$.

M_1 ($\mathrm{M}_\odot$)	$M_{2,\min}$ ($\mathrm{M}_\odot$)	$> 3\,\mathrm{M}_\odot$?	Note
0	1.51	No	Absolute floor
0.5	2.01	No	Extreme low
1.0	2.44	No	Solar-type
2.0	3.16	Yes	Evolved giant
3.0	3.79	Yes	Luminous giant
5.0	4.88	Yes	Supergiant
7.0	5.83	Yes	Bright giant
10.0	7.20	Yes	B-type MS
15.0	9.32	Yes	Early B
20.0	11.3	Yes	O/early B

Table 3. Alternative scenario assessment.

Scenario	Verdict	Key constraint
White dwarf	Excluded	$f(M) = 1.05 \times M_{\mathrm{Ch}}$
Neutron star	Model-dependent	Requires $M_1 \geq 2$
MS companion	Model-dependent	Depends on L_{prim}
Triple system	Unlikely	Circ. orbit + stability
Stripped He star	Disfavoured	No UV data
Artefact	Unlikely	$\sigma = 343$; RUWE 2.32

Table 4. Sensitivity of $P(\mathrm{BH})$ to the M_1 prior. All runs use 2×10^5 Monte Carlo draws per configuration with isotropic inclination priors. The classification shifts from mass-gap to BH as M_1 increases, underscoring the critical dependence on spectroscopic M_1 determination.

Configuration	BH threshold ($\mathrm{M}_\odot$)	M_1 ($\mathrm{M}_\odot$)	σ_{M_1} ($\mathrm{M}_\odot$)	$P(\mathrm{BH})$ (%)	$P(\mathrm{NS})$ (%)	$\tilde{M}_2$ [68% CI] ($\mathrm{M}_\odot$)
Extreme low	5.0	1.0	0.5	34.9	27.1	3.8 [2.7, 11.1]
Cool giant	3.0	2.0	1.0	93.7	6.3	4.9 [3.4, 12.6]
Cool giant	5.0	2.0	1.0	47.8	6.3	4.9 [3.4, 12.6]
Moderate giant	3.0	3.0	1.5	98.5	1.5	5.7 [4.0, 13.9]
Moderate giant	5.0	3.0	1.5	61.9	1.5	5.7 [4.0, 13.9]
Fiducial	3.0	5.0	3.0	99.5	0.5	7.1 [4.7, 16.3]
Fiducial	5.0	5.0	3.0	79.6	0.5	7.1 [4.7, 16.3]
Luminous star	5.0	8.0	3.0	98.5	0.0	8.9 [6.5, 19.1]
B supergiant	5.0	12.0	4.0	100.0	0.0	10.9 [8.2, 22.4]
Pipeline estimate	5.0	15.0	5.0	100.0	0.0	12.3 [9.2, 24.6]